# Materials Design Analysis Reporting (MDAR) Checklist for Authors

The MDAR framework establishes a minimum set of requirements in transparent reporting applicable to studies in the life sciences (see Statement of Task: doi:10.31222/osf.oio/9sm4x.). The MDAR checklist is a tool for authors, editors and others seeking to adopt the MDAR framework for transparent reporting in manuscripts and other outputs. Please refer to the MDAR Elaboration Document for additional context for the MDAR framework.

# Materials

| Antibodies | Yes (indicate where provided: page no/section/legend) | n/a |
|---|---|---|
| For commercial reagents, provide supplier name, catalogue number and RRID, if available | No antibodes detected.<br>Please add identifiers for all resources where possible | |

| Cell Materials | Yes (indicate where provided: page no/section/legend) | n/a |
|---|---|---|
| **Cell lines:** Provide species information, strain. Provide accession number in repository OR supplier name, catalog number, clone number, OR RRID | No cell lines detected<br>Please add identifiers for all resources where possible | |
| **Primary cultures:** Provide species, strain, sex of origin, genetic modification status. | Not currently checked by SciScore | |

| Experimental Animals | Yes (indicate where provided: page no/section/legend) | n/a |
|---|---|---|
| **Laboratory animals:** Provide species, strain, sex, age, genetic modification status. Provide accession number in repository OR supplier name, catalog number, clone number, OR RRID | No organisms detected<br>Please add identifiers for all resources where possible | |
| **Animal observed in or captured from the field:** Provide species, sex and age where possible | Not currently checked by SciScore | |
| **Model organisms:** Provide Accession number in repository (where relevant) OR RRID | See laboratory animals section for information. | |

| Plants and microbes | Yes (indicate where provided: page no/section/legend) | n/a |
|---|---|---|
| **Plants:** provide species and strain, unique accession number if available, and source (including location for collected wild specimens) | Not currently checked by SciScore | |
| **Microbes:** provide species and strain, unique accession number if available, and source | Not currently checked by SciScore | |

| Human research participants | Yes (indicate where provided: page no/section/legend) | n/a |
|---|---|---|
| Identify authority granting ethics approval (IRB or equivalent committee(s), provide reference number for approval. | Not detected. | |
| Provide statement confirming informed consent obtained from study participants. | Not detected. | |
| Report on age and sex for all study participants. | **Age:**, women: 266 age: 51.2 +- 11.4 years ) was used in this study .<br>**Sex:**not detected. | |

# Design

| Study protocol | Yes (indicate where provided: page no/section/legend) | n/a |
|---|---|---|
| For clinical trials, provide the trial registration number OR cite DOI in manuscript. | Not detected. | |

| Laboratory protocol | Yes (indicate where provided: page no/section/legend) | n/a |
|---|---|---|
| Provide DOI or other citation details if detailed step-by-step protocols are available. | Not detected. | |

| Experimental study design (statistics details) | Yes (indicate where provided: page no/section/legend) | n/a |
|---|---|---|
| State whether and how the following have been done, or if they were not carried out | | |
| Sample size determination | not detected. | |
| Randomization | In accordance with the instruction manual, the 9 simulated nodules were randomly attached to the internal structure of the lung by using double-sided adhesive tape and tweezers. | |
| Blinding | not detected. | |
| inclusion/exclusion criteria | However, because of the fixed-shape segmentation, all morphological RFs were excluded. | |

| Sample definition and in-laboratory replication | Yes (indicate where provided: page no/section/legend) | n/a |
|---|---|---|
| State number of times the experiment was replicated in laboratory | For the intra-CT protocol trial, Aquilion ONE TSX-301C was selected to obtain CT images with varying acquisition and reconstruction parameters (Table 2), and the intra-CT protocol trial was repeated twice to obtain a more general result. | |
| Define whether data describe technical or biological replicates | Not detected. | |

| Ethics | Yes (indicate where provided: page no/section/legend) | n/a |
|---|---|---|
| Studies involving human participants: State details of authority granting ethics approval (IRB or equivalent committee(s), provide reference number for approval. | Not detected. | |
| Studies involving experimental animals: State details of authority granting ethics approval (IRB or equivalent committee(s), provide reference number for approval. | Not detected. | |
| Studies involving specimen and field samples: State if relevant permits obtained, provide details of authority approving study; if none were required, explain why. | Not detected. | |

| Dual Use Research of Concern (DURC) | Yes (indicate where provided: page no/section/legend) | n/a |
|---|---|---|
| If study is subject to dual use research of concern, state the authority granting approval and reference number for the regulatory approval | Not currently checked by SciScore | |

# Analysis

| Attrition | Yes (indicate where provided: page no/section/legend) | n/a |
|---|---|---|
| State if sample or data point from the analysis is excluded, and whether the criteria for exclusion were determined and specified in advance. | not required. | |

| Statistics | Yes (indicate where provided: page no/section/legend) | n/a |
|---|---|---|
| Describe statistical tests used and justify choice of tests. | TABLE 1 CT Acquisition Protocols for the Test-Retest and Inter-CT Trials Site CT Scanner (Manufacturer) Pitch* Rotation Time (s) Tube Voltage (kVp) Tube Current (mA*s) FOV (mm) Slice Thickness (mm)* Slice Interval (mm)* Reconstruction Kernel+ Reconstruction Method++ Shanghai Public Health Clinical Center (Jinshan District) Aquilion ONE TSX-301C (Toshiba) 0.813 0.5 120 75 350 1.0 1.0 FC56 AIDR 3D (standard) Shanghai Public Health Clinical Center (Jinshan District) SCENARIA (Hitachi) 0.8281 0.5 120 75 350 1.0 1.0 66 Intelli IP (Lv.2) Shanghai Public Health Clinical Center (Hongkou District) Brilliance 64 (Philips) 0.891 0.5 120 75 350 1.0 1.0 L Standard (enhancement = 1.0) Zhongshan Hospital of Fudan University SOMATOM Definition AS (Siemens Healthineers) 0.9 0.5 120 75 350 1.0 1.0 B60f ADMIRE (strength = 3) Zhongshan Hospital of Fudan University Aquilion ONE TSX-301A (Toshiba) 0.828 0.5 120 75 350 1.0 1.0 FC56 AIDR 3D (standard) Zhongshan Hospital of Fudan University UCT550 (United Imaging Healthcare) 0.8875 0.5 120 75 350 1.0 1.0 SHARPC Adaptive filter function (enhancement = 2.5) Shanghai Sixth Peoples Hospital Revolution CT (GE Healthcare) 0.992 0.5 120 75 350 1.25 1.25 LUNG ASiR (PlusSS40) The number of decimal places was consistent with the corresponding CT scanner control panel. | |

| Data availability | Yes (indicate where provided: page no/section/legend) | n/a |
|---|---|---|
| State whether newly created datasets are available, including protocols for access or restriction on access. | Not detected. | |
| If data are publicly available, provide accession number in repository or DOI or URL. | The varying scope of acquisition and reconstruction parameters were chosen based on previous studies 16,18,19,28,29 and radiologists previous knowledge. | |
| If publicly available data are reused, provide accession number in repository or DOI or URL, where possible. | Not detected. | |

| Code availability | Yes (indicate where provided: page no/section/legend) | n/a |
|---|---|---|
| For all newly generated code and software essential for replicating the main findings of the study: | | |
| State whether the code or software is available. | Not detected. | |
| If code is publicly available, provide accession number in repository, or DOI or URL. | Not detected. | |

## Analysis

| Adherence to community standards | Yes (indicate where provided: page no/section/legend) | n/a |
|---|---|---|
| MDAR framework recommends adoption of discipline-specific guidelines, established and endorsed through community initiatives. Journals have their own policy about requiring specific guidelines and recommendations to complement MDAR. | | |
| State if relevant guidelines (eg., ICMJE, MIBBI, ARRIVE) have been followed, and whether a checklist (eg., CONSORT, PRISMA, ARRIVE) is provided with the manuscript. | Not currently checked by SciScore | |